

Lecture 7: By Force of Habit

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Intro

- Part I: derive the Campbell-Cochrane habit model ([Campbell and Cochrane 1999](#))
- Part II: calibration and quantitative discipline
- Part III: follow-up tracks in equity, bonds, and FX
- Core state variable: surplus consumption ratio, S_t

Part I: Let's Derive the Model

Why Habit?

- Time-separable CRRA struggles with high equity premia and low/steady r_f ([Mehra and Prescott 1985](#); [Weil 1989](#))
- We want countercyclical risk prices without implausible interest-rate dynamics
- External habit creates a recession state variable
- Target: high Sharpe ratios in bad times

- Representative agent utility with external habit:

$$U_t = \mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j \frac{(C_{t+j} - X_{t+j})^{1-\gamma} - 1}{1-\gamma} \right]$$

- X_t : habit, C_t : consumption, β : discount factor, γ : power parameter

- Define the surplus consumption ratio and its log state:

$$S_t \equiv \frac{C_t - X_t}{C_t}, \quad s_t \equiv \log S_t$$

- Local curvature is state dependent: $\eta_t = \gamma/S_t$
- Low S_t (consumption near habit) means high effective risk aversion

- Campbell-Cochrane law of motion:

$$s_{t+1} = (1 - \phi)\bar{s} + \phi s_t + \lambda(s_t)(\Delta c_{t+1} - g), \quad \Delta c_{t+1} = g + \sigma \varepsilon_{t+1}$$

- ϕ : persistence, g : mean consumption growth, $\lambda(s_t)$: sensitivity function

- Intertemporal marginal rate of substitution:

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \left(\frac{S_{t+1}}{S_t} \right)^{-\gamma}$$

- Risk premia move through both consumption growth and surplus-consumption dynamics

- Maximum conditional Sharpe ratio is pinned down by SDF volatility:

$$\max_R \frac{\mathbb{E}_t[R_{t+1}^e]}{\sigma_t(R_{t+1}^e)} = \sqrt{\exp(\gamma^2 \sigma^2 [1 + \lambda(s_t)]^2) - 1}$$

- In recessions (s_t low), $\lambda(s_t)$ is high, so Sharpe ratios rise

- Implied real risk-free rate:

$$r_{f,t} = -\log(\beta) + \gamma g - \gamma(1 - \phi)(s_t - \bar{s}) - \frac{\gamma^2 \sigma^2}{2} [1 + \lambda(s_t)]^2$$

- Intertemporal-substitution and precautionary-savings terms offset by construction

- Functional-form choice in Campbell and Cochrane (1999):

$$\lambda(s_t) = \begin{cases} \frac{1}{\bar{S}} \sqrt{1 - 2(s_t - \bar{s})} - 1, & s_t \leq s_{\max} \\ 0, & s_t > s_{\max} \end{cases}$$

- Delivers: near-steady predetermined habit, nonnegative habit response, stable r_f

- Bad times: S_t low $\Rightarrow$ marginal utility highly sensitive
- Price of risk rises exactly when cash flows are weak
- Expected returns become countercyclical
- Valuation ratios fall in recessions

TODO Figure: Sensitivity Function + Habit Response

- TODO: insert Figure 1 from Campbell and Cochrane (1999)
- Intended exhibits: $\lambda(s_t)$ and $d \log X_t / d \log C_t$

Questions?

Part II: Calibration

- Baseline sample in Campbell and Cochrane (1999): postwar U.S. consumption and equity data
- Match consumption moments: g, σ
- Match valuation-ratio persistence: ϕ
- Match risk-free rate and Sharpe ratio with β and γ

- Annualized baseline choices:
- $g = 1.89\%$, $\sigma = 1.50\%$, $\phi = 0.87$, $\gamma = 2.0$
- Dividend process inputs: $\sigma_d = 11.2\%$, $\rho(\Delta d, \Delta c) = 0.2$
- Implied: $\beta \approx 0.89$, $\bar{S} \approx 0.057$, $S_{\max} \approx 0.094$

TODO Table: Campbell-Cochrane Table 1

- TODO: insert calibration table from Campbell and Cochrane ([1999](#)) (Table 1)
- Intended exhibit: assumed and implied parameters

- The model targets: equity premium, Sharpe ratio, and low average r_f
- It reproduces high return volatility with smooth consumption growth
- It generates persistent valuation-ratio dynamics and return predictability
- These are out-of-sample moments relative to calibration targets

TODO Table: Campbell-Cochrane Table 2

- TODO: insert means and standard deviations from Campbell and Cochrane ([1999](#)) (Table 2)
- Intended exhibit: model vs data moments

- Low S_t predicts high expected returns (equity and consumption claims)
- Return volatility is countercyclical and persistent
- Long-horizon predictability rises with horizon
- Volatility-test decompositions load heavily on discount-rate variation

TODO Figure/Table: Campbell-Cochrane Figures 4-6 / Tables 5-6

- TODO: insert Figure 4 (expected returns), Figure 5 (volatility), Figure 6 (Sharpe ratios)
- TODO: insert Tables 5-6 (long-horizon predictability and variance decomposition)

Questions?

Part III: Where Habit Models Went

Three Follow-Up Tracks

- Track 1 (equity): richer predictability and valuation implications
- Track 2 (term structure): bond premia under habit-based discount rates
- Track 3 (international/FX): currency premia from cross-country habit risk
- Same backbone as Parts I-II: state-dependent SDF via S_t

- Extends the habit mechanism to a broader equity-pricing setting ([Menzly, Santos, and Veronesi 2004](#))
- Keeps countercyclical risk prices driven by the surplus-consumption state
- Uses the same discount-rate channel to organize predictability facts
- Direct link: Part I mechanism + Part II discipline, but richer equity objects

TODO Figure/Table: MSV (2004)

- TODO: insert one key empirical figure from Menzly, Santos, and Veronesi ([2004](#))
- TODO: insert one key table (moments/predictability fit)

- Brings the habit state variable to bond pricing ([Wachter 2006](#))
- Term premia become state dependent through the same risk-price channel
- Generates a joint view of expected stock and bond returns
- Direct link: same S_t -driven discount-rate variation, new asset class

TODO Figure/Table: Wachter (2006)

- TODO: insert one yield-curve or bond-risk-premium figure from Wachter ([2006](#))
- TODO: insert one table with term-structure moments/predictability

- Two-country habit framework for currency risk premia ([Verdelhan 2010](#))
- Carry returns load on cross-country differences in bad-times risk
- Habit-based SDFs rationalize the forward-premium puzzle patterns
- Direct link: same mechanism from Parts I-II, exported to FX markets

TODO Figure/Table: Verdelhan (2010)

- TODO: insert one key FX/carry figure from Verdelhan ([2010](#))
- TODO: insert one table with currency-risk-premium evidence

- Common mechanism: S_t controls local curvature and the price of risk
- Calibration logic: match core moments, then test dynamic implications
- Main difference: which assets load most on recession-state risk
- Habit is a transportable discount-rate engine across markets

TODO Comparative Table: Mechanism vs Data vs Limits

- TODO: insert comparison table for Menzly, Santos, and Veronesi ([2004](#)), Wachter ([2006](#)), Verdelhan ([2010](#))
- Intended columns: mechanism, targeted facts, main success, open limitation

- External habit resolves key aggregate puzzles through state-dependent risk prices
- Calibration discipline is essential: moments first, dynamics second
- Follow-ups show the same mechanism in equities, bonds, and FX
- Next step in the course: compare habits to long-run risks and disasters

Questions?

- Campbell and Cochrane (1999): external habit makes risk prices countercyclical
- Surplus consumption ratio S_t is the key state variable
- Calibration matches equity premium facts with a low, stable risk-free rate
- Follow-ups map the same mechanism into equities, bonds, and FX

Readings and References i

- Campbell and Cochrane (1999)
- Menzly, Santos, and Veronesi (2004)
- Wachter (2006)
- Verdelhan (2010)
- Background: Mehra and Prescott (1985), Weil (1989)

Campbell, John Y., and John H. Cochrane. 1999. "By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior." *Journal of Political Economy* 107 (2): 205–51.
<https://doi.org/10.1086/250059>.

Mehra, Rajnish, and Edward C. Prescott. 1985. "The Equity Premium: A Puzzle." *Journal of Monetary Economics* 15 (2): 145–61. [https://doi.org/10.1016/0304-3932\(85\)90061-3](https://doi.org/10.1016/0304-3932(85)90061-3).

- Menzly, Lior, Tano Santos, and Pietro Veronesi. 2004. "Understanding Predictability." *Journal of Political Economy* 112 (1): 1–47. <https://doi.org/10.1086/379934>.
- Verdelhan, Adrien. 2010. "A Habit-Based Explanation of the Exchange Rate Risk Premium." *The Journal of Finance* 65 (1): 123–46. <https://doi.org/10.1111/j.1540-6261.2009.01525.x>.
- Wachter, Jessica A. 2006. "A Consumption-Based Model of the Term Structure of Interest Rates." *Journal of Financial Economics* 79 (2): 365–99. <https://doi.org/10.1016/j.jfineco.2005.02.004>.
- Weil, Philippe. 1989. "The Equity Premium Puzzle and the Risk-Free Rate Puzzle." *Journal of Monetary Economics* 24 (3): 401–21. [https://doi.org/10.1016/0304-3932\(89\)90028-7](https://doi.org/10.1016/0304-3932(89)90028-7).